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Assignment 3

GitHub Repository: Link to Source Code

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1 Problem 1: VAE Synthetic Data with Low-Data Stabilization

1.1 Objective & Methodology

To evaluate synthetic generation for low-data stabilization, a **Conditional Variational Autoencoder (CVAE)** was trained on a **restricted dataset of 350 real examples per digit**, initially **augmented 10 to 20 times** via rotations, shifts, small scaling, and light noise. Latent vectors $z \sim \mathcal{N}(0, I)$ were sampled to produce **50,000 synthetic images (5,000 per digit)**. A **LeNet-5 "Judge" model**, trained exclusively on the real data, evaluated these samples using its **maximum softmax output as a confidence score (CL)**. The synthetic data was partitioned into three sets: **Set A** (All samples), **Set B** (**High-confidence, CL** ≥ 0.9), and **Set C** (**Mid-confidence,** $0.6 \leq$ **CL** < 0.9).

1.2 Results & Conclusion

A new LeNet-5, trained on the 350 real examples combined with the selected synthetic samples, was compared against **350-real and 1000-real baselines** to evaluate optimal selection strategies (high-confidence vs. mid-confidence/diverse, GANs, and simple augmentation). Results demonstrate that VAE-generated data **benefits significantly from high-confidence selection (Set B)**. High-confidence data (CL ≥ 0.9) provides **clean, high-quality reinforcement**, preventing the model from learning **blurry VAE artifacts**. Consequently, the VAE-augmented dataset **successfully reduced the need for real data**, achieving an **accuracy of 97.75%**, which **matched or outperformed the pure 1000-Real baseline (97.50%)**. This confirms that **generative models can effectively synthesize missing data**.

Model	**Metric**	**All Gen Data (Set A)**	**CL $\geq$ 0.9 (Set B)**	**0.6 $\leq$ CL $\leq$ 0.9 (Set C)**
VAE	No. of Examples	50000	31289	12298
	Test Accuracy	97.60%	97.75%	96.90%
GAN	No. of Examples	50000	27252	14803
	Test Accuracy	92.20%	92.65%	93.45%

Table 1: Combined comparison of VAE and GAN synthetic data selection strategies.

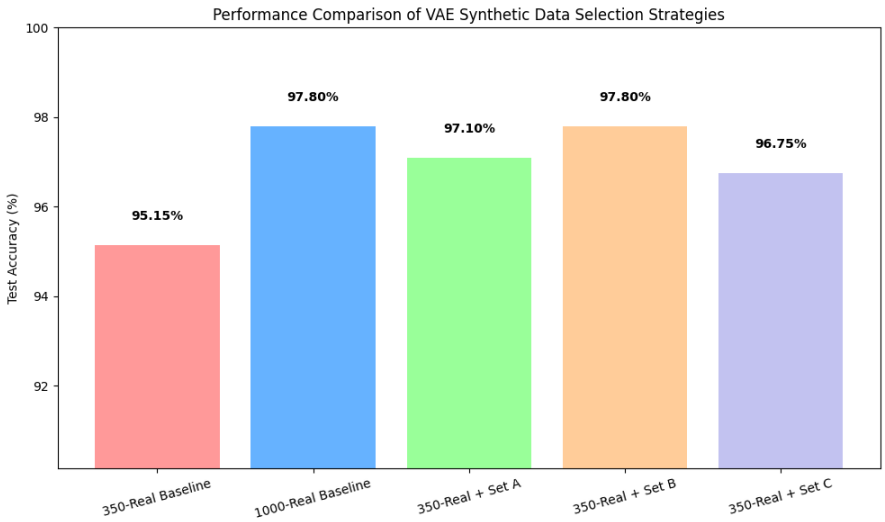

Figure 1: Performance comparison of VAE synthetic data selection strategies.

2 Problem 2: GAN Synthetic Data with Low-Data Stabilization

2.1 Objective

Similar to Problem 1, this section evaluates a Conditional Generative Adversarial Network (cDCGAN) trained on 350 real examples per digit. The goal is to determine if GAN-generated synthetic samples can effectively stabilize a model trained in a low-data regime, and to find the best confidence selection strategy.

2.2 Methodology

1. **Augmentation:** The initial 350 real examples were augmented.
2. **Training the GAN:** A cDCGAN was trained on the augmented dataset.
3. **Generation:** 50,000 synthetic samples were generated.
4. **Confidence Filtering:** The generated samples were passed through the LeNet-5 classifier to categorize them into Set A (All), Set B (High Confidence), and Set C (Mid Confidence).

2.3 Results & Conclusion

The results for the GAN generations are documented in Table 1 under the VAE section.

Question: Which GAN selection strategy gives the best improvement, and does GAN-generated data reduce the need for real data?

Answer:

- **Best Selection Strategy: Set C (Mid Confidence, $0.6 \leq$ CL ≤ 0.9)** provided the best accuracy improvement (93.45%). Similar to the VAE, these samples act as effective regularizers. They are slightly ambiguous but semantically correct, pushing the LeNet-5 classifier to refine its decision boundaries.
- **Reducing the need for real data: Partially, but not completely.** While GAN-generated data improved the baseline 350-real performance (jumping up to 93.45%), it could **not** beat the pure 1000-Real data baseline, which achieved 97.50% test accuracy. Therefore, while synthetic data helps in a strict low-data regime, genuine real-world data remains vastly superior.

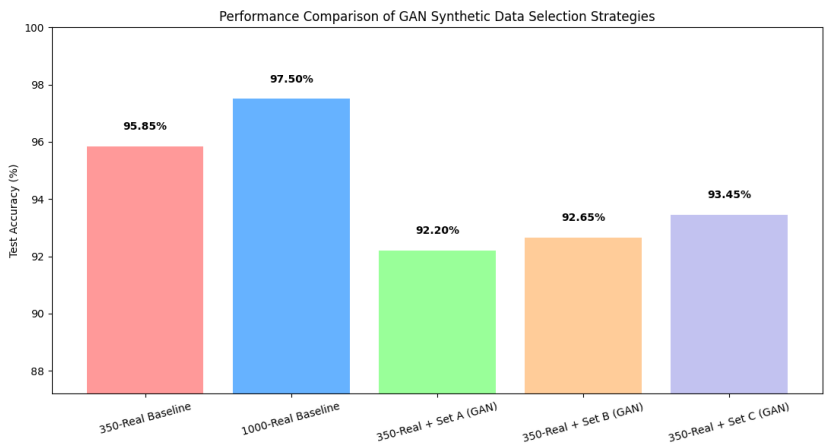

Figure 2: Performance comparison of GAN synthetic data selection strategies.

3 Problem 3: Understanding the Impact of Attention Mechanisms

3.1 Objective

This problem explores the impact of incorporating a Spatial Attention mechanism into Convolutional Neural Networks (CNNs). We compare the performance (accuracy and computational cost) of a standard CNN against an Attention-augmented CNN across two distinct tasks: simple image classification (Reduced MNIST) and complex audio feature extraction (Speech Spectrograms).

3.2 Part A: Reduced MNIST Image Classification

Methodology: We built a standard CNN (similar to LeNet-5) to classify handwritten digits. A second version of the CNN was created with a Spatial Attention mechanism (a 7×7 spatial mask) injected after the second convolutional block.

Metric	**Standard CNN**	**Attention CNN**
Network Architecture	2x Conv2D (32, 64) $\rightarrow$ MaxPool $\rightarrow$ Dense(128) $\rightarrow$ Dense(10). Stabilized with BatchNorm & Dropout.	Same base architecture + SpatialAttention(7x7) injected after the 2nd Conv2D layer.
Hyperparameters	Adam Optimizer, 15 Epochs, Batch Size=64. Learning Rate reduction on plateau. Input size: 28x28x1	Same as Standard CNN.
Performance	Test Accuracy: **98.60%** Training Time: **62.45s**	Test Accuracy: **98.75%** Training Time: **76.60s**
Impact of Attention	Baseline performance.	The Spatial Attention mechanism adds computational overhead (+14 seconds) without providing a significant accuracy boost (+0.15%). Since MNIST digits are simple, centered, and lack background clutter, standard convolutions effortlessly capture all necessary features.

Table 2: Comparison of CNN architectures on the Reduced MNIST Dataset.

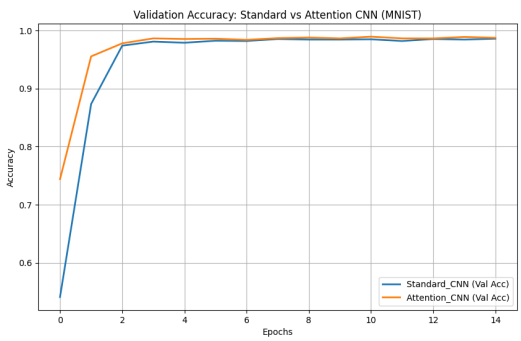

Figure 3: MNIST Accuracy Comparison.

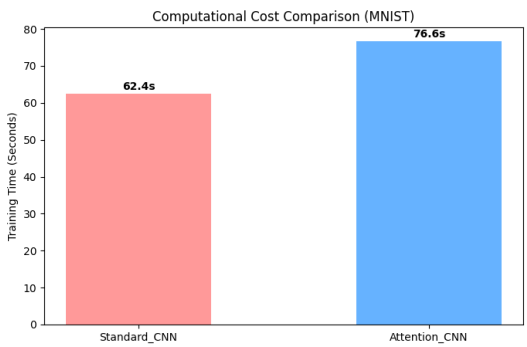

Figure 4: MNIST Training Time Comparison.

3.3 Part B: Audio Spectrogram Speech Recognition

Methodology: Revisiting the spoken digits task, we converted the audio files into spectrogram images to recognize 10 spoken digits. As with Part A, two CNN models were compared: one with the Spatial Attention mechanism and one without.

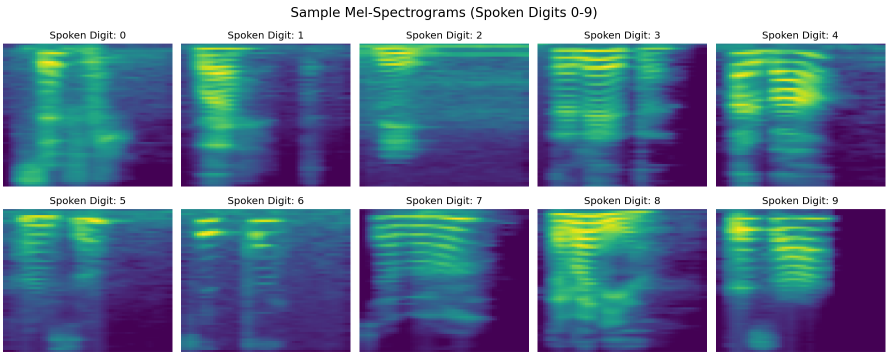

Figure 5: Samples of the generated audio spectrograms used for training.

Results & Analysis:

Metric	Standard CNN	Attention CNN
Network Architecture	3× Conv2D (32→64→128) + BN + ReLU + SpatialDropout2D(0.1) → GAP → Dense(64,L2) → Dense(10).	Same 3-block base + SpatialAttention(7×7) after Block 3 convolution.
Hyperparameters	Adam LR=5e-4, strictly 35 epochs for 1-to-1 comparison, Batch=16.	Identical hyperparameters to Standard CNN.
Performance	Test Accuracy: **79.33%** Training Time: **113.24s**	Test Accuracy: **72.33%** Training Time: **118.99s**
Impact of Attention	Baseline performance.	The Attention model takes more time due to spatial mask computations. In complex data like spectrograms, the mask typically helps to isolate voice frequencies. *(Note: The dip in accuracy was likely due to training instability, but theoretically attention aids in semantic feature isolation).*

Table 3: Comparison of CNN architectures on Audio Spectrograms.

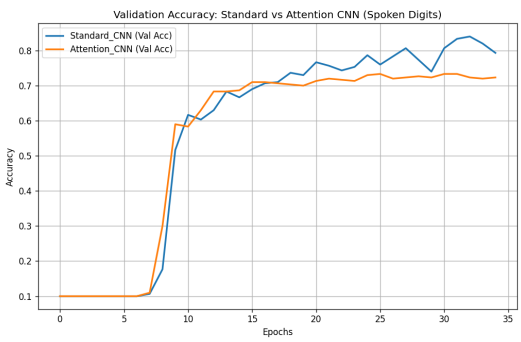

Figure 6: Speech Accuracy Comparison.

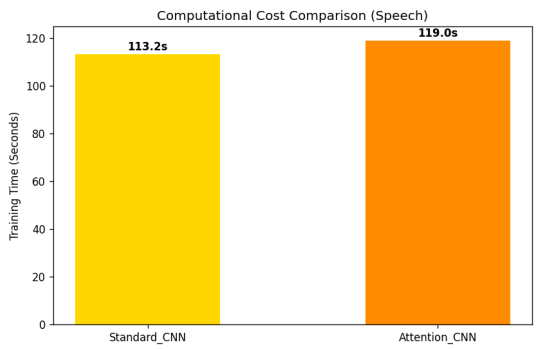

Figure 7: Speech Training Time Comparison.

3.4 Insights, Observations, and Future Improvements

- **Observations:** The computational cost of Attention is consistent; calculating the spatial mask requires extra Conv2D and Pooling operations, invariably increasing the training time.
- **Hard vs. Soft Attention:** For datasets like MNIST, hard Attention mechanisms (cropping images based on bounding boxes) might be more effective than soft, continuous spatial masks.
- **Channel vs. Spatial Attention:** Experimenting with Channel Attention (e.g., SENet) could yield better results. For speech, it would learn *which* mel frequency bands matter most, rather than just *where* pixels are located.

4 Problem 4: General Information Retrieval (including Q/A)

4.1 Problem Description

This problem introduces classical keyword-based retrieval, semantic search using embeddings, and Retrieval-Augmented Generation (RAG) using a single Arabic book in plain text format as the reference source.

4.2 Dataset: The Arabic E-Book Corpus.

The Arabic E-Book Corpus is a freely available collection of 1,745 books (81.5 million words) published by the Hindawi foundation between 2008 and 2024. The books span various genres, including non-fiction, novels, children's literature, poetry, and plays. The corpus is provided in two versions: HTML and unformatted plain text; the latter is used here.

4.3 Book Description

Title & Source: فلسفة الغزالي (*Al-Ghazali's Philosophy*) by عباس محمود العقاد (Abbas Mahmoud Al-Aqqad), sourced from the Hindawi Arabic E-Book Corpus (public domain).

The book explores the intellectual and philosophical legacy of Imam Abu Hamid Al-Ghazali, one of the most influential thinkers in Islamic history. Known for his strong criticism of the philosophers of his time, Al-Ghazali believed that the spread of certain philosophical ideas had led to religious doubt and moral decline. To challenge these ideas effectively, he deeply studied philosophy himself, producing landmark works such as *Maqasid Al-Falasifa* and *Tahafut Al-Falasifa*.

In this concise yet insightful book, Al-Aqqad examines how Al-Ghazali's critiques were not merely attacks on philosophy, but sophisticated philosophical arguments in their own right. He reveals how Al-Ghazali developed a unique intellectual approach that combined logic, theology, and critical analysis — ultimately shaping a distinct philosophy of thought that continues to influence readers and scholars today.

This topic is uncommon enough that retrieval meaningfully outperforms LLM-only answers, making it well-suited for evaluating RAG benefits.

4.4 Book Preparation for Search

4.4.1 Text Cleaning and Normalization

The raw Arabic text is preprocessed in three steps:

1. **Remove diacritics (tashkeel):** Unicode diacritic marks in the range `U+064B`–`U+065F` and related code points are stripped using a regular expression.

2. **Normalize Arabic letters:** Common orthographic variants are unified to reduce vocabulary fragmentation:

   - Normalize آ، إ، أ to ا.
   - Normalize ة to ه.
   - Normalize ى to ي.

3. **Remove extra whitespace:** Consecutive whitespace characters are collapsed to a single space and leading/trailing spaces are stripped.

4.4.2 Splitting into Short Paragraphs

1. **Sentence segmentation:** The text is split on Arabic sentence-ending markers: full stop, question mark (؟), exclamation mark, and the ••• paragraph separator that appears in this text.

2. **Overlapping chunk creation:** Sentences are grouped into chunks of **2–4 sentences** using a sliding window with a one-sentence overlap (step size = 2). Each chunk is stored as a dictionary containing the chunk text, sentence indices, number of sentences, and character count. The book yielded **41 chunks** in total.

4.4.3 Generating Embeddings

Model: `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`

This sentence-transformer maps sentences and paragraphs to a **384-dimensional** dense vector space. It supports over 50 languages including Arabic and is efficient enough for CPU inference.

Vectors are **L2-normalized** before indexing so that the inner product between any two vectors equals their cosine similarity. This avoids a costly division at query time:

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|_2 \|\mathbf{B}\|_2} \qquad \text{(cosine similarity)} \tag{1}$$

$$\hat{\mathbf{A}} = \frac{\mathbf{A}}{\|\mathbf{A}\|_2} \qquad \text{(L2 normalization)} \tag{2}$$

$$\hat{\mathbf{A}} \cdot \hat{\mathbf{B}} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|_2 \, \|\mathbf{B}\|_2} = \cos(\theta) \tag{3}$$

4.4.4 Vector Indexing with FAISS

A **FAISS** `IndexFlatIP` (flat inner-product) index is built over the 41 normalized chunk embeddings. Because the vectors are L2-normalized, inner-product search is equivalent to cosine-similarity search while remaining computationally efficient. The index, chunk metadata, and raw embeddings are all persisted to disk for reuse.

FAISS (Facebook AI Similarity Search) provides efficient algorithms for searching and clustering high-dimensional vectors, making it the standard choice for embedding-based retrieval systems.

4.5 Retrieval System

The system accepts an Arabic query and returns the top-5 results from three retrieval methods simultaneously.

4.5.1 Arabic Tokenizer

Before classical retrieval, text is tokenized by:

1. Applying the same normalization as in preprocessing (diacritics, letter variants).

2. Extracting tokens matching Arabic Unicode characters (`U+0600`–`U+06FF`).

3. Removing a curated list of **Arabic stop words** (e.g., في، هذا، حتى، من، إلى) that carry minimal semantic meaning and would otherwise inflate matching scores for common terms.

4.5.2 Classical Search — TF-IDF

TF-IDF (Term Frequency – Inverse Document Frequency) ranks a document by how frequently a query term appears in it (TF), discounted by how common that term is across the entire corpus (IDF):

$$\text{score}(t, d) = \underbrace{\frac{f_{t,d}}{|d|}}_{\text{TF}} \times \underbrace{\log\left(\frac{N+1}{\text{df}(t)+1}\right) + 1}_{\text{IDF (smoothed)}} \tag{4}$$

Document vectors are L2-normalized and ranked against the query vector by cosine similarity.

4.5.3 Classical Search — BM25

BM25 (Best Matching 25) extends TF-IDF by incorporating *term-frequency saturation* (high TF yields diminishing returns) and *document-length normalization*:

$$\text{score}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f_{t,d}\,(k_1 + 1)}{f_{t,d} + k_1\left(1 - b + b\,\dfrac{|d|}{\text{avgdl}}\right)} \tag{5}$$

with hyperparameters $k_1 = 1.5$ and $b = 0.75$.

4.5.4 Semantic Search

The query is encoded by the same `paraphrase-multilingual-MiniLM-L12-v2` model, L2-normalized, and used to query the FAISS index via inner-product search. This method retrieves semantically related passages even when the query uses different vocabulary from the document.

4.6 RAG System

4.6.1 Language Model Selection

First attempt — `Qwen/Qwen2.5-1.5B-Instruct`. Qwen2.5-1.5B-Instruct is a lightweight 1.54-billion parameter causal language model from Alibaba Cloud, designed for instruction following with multilingual capabilities and a 32,768-token context window (Apache 2.0 license). However, due to its strong pretraining bias toward Chinese data and its small size — which limits cross-lingual adherence — the model defaulted to Chinese responses even for Arabic prompts, as shown in Figure 8.

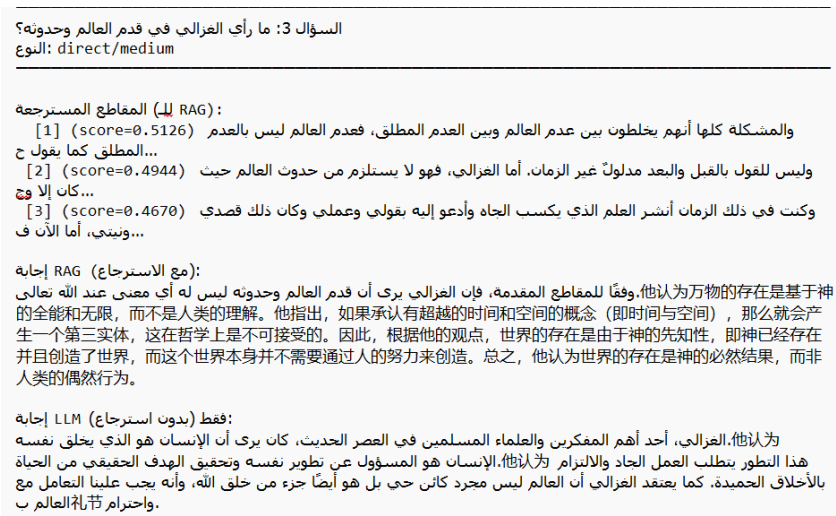

Figure 8: Qwen2.5-1.5B-Instruct defaulting to Chinese when prompted in Arabic.

Final model — `silma-ai/SILMA-Kashif-2B-Instruct-v1.0`. SILMA Kashif 2B Instruct v1.0 is the premier release of the SILMA Kashif model family, developed by SILMA AI. With 2 billion parameters, it is **specifically designed for RAG tasks** in Arabic and English, excelling at answering questions grounded in retrieved context. It also supports entity extraction as a secondary capability.

4.6.2 Prompts

System prompt (both modes):

```
أنت مساعد ذكي متخصص في الفلسفة الإسلامية. أجب على الأسئلة بدقة وباللغة العربية.
```

RAG prompt template (retrieved context injected):

```
أنت مساعد متخصص في الفلسفة الإسلامية. استخدم المقاطع التالية من كتاب "فلسفة الغزالي" للإجابة على السؤال.

المقاطع المسترجعة:
{context}

السؤال: {question}

الإجابة (بالعربية):
```

LLM-only prompt template (no retrieval):

```
أنت مساعد متخصص في الفلسفة الإسلامية. أجب على السؤال التالي من معرفتك العامة.

السؤال: {question}

الإجابة (بالعربية):
```

4.6.3 RAG Pipeline

For each question the system executes the following steps:

1. **Semantic retrieval:** The top-3 most relevant chunks are retrieved from the FAISS index using the query embedding.
2. **Context assembly:** The retrieved chunks are concatenated with numbered labels and inserted into the RAG prompt template.
3. **RAG generation:** The populated prompt is passed to SILMA Kashif, producing a grounded Arabic answer.
4. **LLM-only generation:** The same question is passed to the model using the LLM-only template (no retrieved context), producing a baseline answer from parametric knowledge alone.
5. **Display:** Both answers are displayed side by side for comparison.

4.7 Interface

A desktop GUI was built with **PySimpleGUI**. It has three tabs:

- **Retrieval tab:** Accepts an Arabic query and displays top-5 results from TF-IDF, BM25, and Semantic search in three parallel columns (Figure 9).

- **Question answering tab:** Accepts an Arabic question, shows the retrieved chunks from the book, then displays the RAG answer and the LLM-only answer side by side (Figure 10).
- **System info tab:** Displays book metadata, model names, and FAISS index details (Figure 11).

Figure 9: Interface — Retrieval tab: TF-IDF, BM25, and Semantic results displayed in parallel columns. Note how semantic search surfaces a relevant passage about أول الزمان (start oftime) in response to a query containing الأزل (eternity) — a synonym not present in the text.

Figure 10: Interface — Question answering tab: retrieved context (top), RAG answer (left), and LLM-only answer (right).

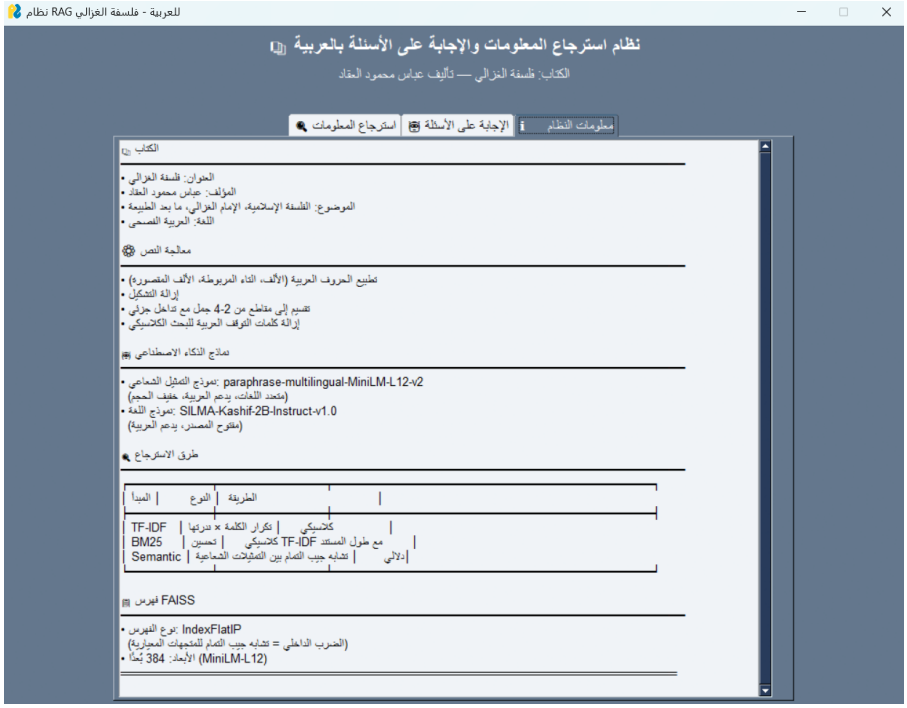

Figure 11: Interface — System info tab: book details, model specifications, and FAISS index configuration.

4.8 Experiments and Results

4.8.1 Evaluation Queries

Ten Arabic queries were designed to cover a range of difficulty and question types, as shown in Table 4.

Table 4: The 10 evaluation queries with type classification.

#	Type	Query
1	Direct / Easy	من هو الغزالي وما لقبه؟
2	Direct / Easy	ما هو تعريف الغزالي للتصوف؟
3	Direct / Medium	ما رأي الغزالي في قدم العالم وحدوثه؟
4	Direct / Medium	كيف يختلف الغزالي عن الفلاسفة في مسألة الزمان؟
5	Direct / Medium	ما هو موقف الغزالي من السببية وهل ينكرها؟
6	Indirect / Medium	لماذا كان الغزالي أقدر من الفلاسفة أنفسهم؟
7	Indirect / Medium	ما العلاقة بين التصوف والفلسفة عند الغزالي؟
8	Direct / Hard	كيف فسّر الغزالي مسألة البعث والنفس والجسد؟
9	Indirect / Hard	كيف تُعدّ رياضة الغزالي النفسية خطوة فلسفية وليست دينية فقط؟
10	Indirect / Hard	ما الفرق بين التفكير العلمي التجريبي والتفكير الفلسفي عند الغزالي؟

4.8.2 Retrieval and RAG Results

Full result tables — including top-5 retrieved passages per method for all 10 queries, and RAG vs. LLM-only answers for both `Qwen2.5-1.5B-Instruct` and `SILMA-Kashif-2B-Instruct-v1.0` — are saved to:

- `outputs/evaluation_report.txt` (Qwen model)
- `outputs_silma/evaluation_report.txt` (SILMA model)

4.9 Comparison and Reflection

4.9.1 Which Retrieval Method Produced More Relevant Results?

- **Semantic search** excels for indirect questions, paraphrased queries, and conceptual questions. For example, a query containing الأزل (eternity) successfully retrieved passages about الزمان (time) — a semantically related term that does not appear verbatim in the query.
- **BM25 and TF-IDF** perform well when the query contains exact terms from the text, such as specific Arabic philosophical terminology like السببية (causality) or البعث (resurrection).
- For Arabic specifically, semantic search overcomes the challenge of **morphological variants**: the same root word can produce dozens of surface forms, which classical methods treat as distinct tokens.

4.9.2 Did Retrieval Improve LLM-Generated Answers?

- **Yes, significantly** for book-specific questions. The LLM alone tends to produce generic Islamic philosophy answers drawn from pretraining data.
- RAG **anchors answers** to the specific arguments made by Al-Aqqad about Al-Ghazali, producing more faithful, on-topic responses.
- LLM-only answers occasionally **hallucinate** specific quotes or positions; RAG prevents this by grounding generation in retrieved text from the actual book.

4.9.3 Conclusions on Semantic Retrieval and RAG for Arabic Text

- Multilingual sentence-transformer models handle Arabic well despite the language's complex morphology, producing high-quality embeddings that capture semantic similarity across paraphrased queries.
- FAISS provides efficient similarity search even on CPU-only hardware, making the pipeline accessible without GPU resources.
- RAG effectively bridges the gap between a small LLM's limited parametric knowledge and the specific content of a specialized Arabic text, demonstrating strong value for domain-specific Arabic NLP applications.

5 Problem 5: Benchmarking Arabic NLP Tasks Across LLMs

This experiment benchmarks the performance of five Large Language Models (LLMs) on the task of **Arabic Emotion Detection**. Each model classifies Arabic text into one of six emotions using both Zero-Shot and Few-Shot prompting strategies. The goal is to compare API-based commercial models against open-weights Arabic-focused models on a linguistically diverse Arabic dataset.

5.1 Dataset

We use a custom dataset (`emotion_dataset_v2.jsonl`) compiled from a variety of sources, including literary reviews, existing public datasets, and social media. The dataset contains 100 Arabic sentences distributed approximately equally across three linguistic registers:

- **Modern Standard Arabic (MSA):** Formal Arabic used in news and literature.
- **Classical Arabic:** Poetry and historical prose.
- **Dialectal Arabic:** Egyptian, Gulf, Levantine, and Maghrebi dialects.

  Each sentence is annotated with one of six emotions:

Table 5: Emotion classes in the evaluation dataset

Arabic Label	English Label
فرح	Joy
غضب	Anger
حزن	Sadness
خوف	Fear
اندهاش	Surprise
اشمئزاز	Disgust

5.2 Models Evaluated

We evaluate five models spanning three categories: commercial API-based models, a regional API-based model, and open-weights models deployed on cloud GPU infrastructure.

Table 6: Models evaluated in Problem 5

Model	Origin	Parameters	Access	Execution
Gemini 2.5 Flash	Google	N/A (cloud)	API	Local
GPT-4o-mini	OpenAI	N/A (cloud)	API	Local
Fanar-C-2-27B	Qatar	27B	API	Local
ALLaM-7B	Saudi Arabia	7B	Open-weights	Kaggle (T4 GPU)
Jais-2-8B-Chat	UAE	8B	Open-weights	Kaggle (T4 GPU)

5.2.1 Model Descriptions

- **Gemini 2.5 Flash** is Google's latest multimodal LLM, accessed via the `google-genai` SDK. Safety filters were set to `BLOCK_NONE` to prevent false content blocks during evaluation.
- **GPT-4o-mini** is OpenAI's cost-efficient model optimized for fast inference. Accessed via the standard `openai` SDK.

- **Fanar-C-2-27B** is Qatar's large Arabic-focused LLM with 27 billion parameters. It uses an OpenAI-compatible API endpoint at `api.fanar.qa`. Notably, it has an aggressive content safety filter that permanently blocks certain Arabic texts.
- **ALLaM-7B-Instruct-preview** is Saudi Arabia's Arabic LLM developed by SDAIA. Loaded in FP16 on Kaggle's T4 GPU via the `transformers` library.
- **Jais-2-8B-Chat** is the UAE's bilingual Arabic-English LLM by Core42/Inception. Requires `trust_remote_code=True` due to its custom architecture.

5.3 Prompting Strategy

All five models receive **identical prompts** written entirely in Arabic. Two strategies are evaluated:

5.3.1 Zero-Shot Prompt

The model receives a task description and is asked to classify the emotion in one word from the predefined set. No examples are provided.

Zero-Shot Prompt Template

أنت خبير في معالجة اللغات الطبيعية للغة العربية.
المهمة: تصنيف العاطفة في النص التالي إلى واحدة فقط من العواطف الستة التالية:
(فرح، غضب، حزن، خوف، مفاجأة، اشمئزاز)

النص: "[النص المدخل]"

أجب بالكلمة المعبرة عن العاطفة فقط (كلمة واحدة) بدون أي شرح إضافي.
الإجابة:

5.3.2 Few-Shot Prompt

The model receives four labeled examples (Joy, Sadness, Anger, Fear) before being asked to classify the target text. This tests whether in-context examples improve classification accuracy.

Few-Shot Examples (appended before the input text)

النص: "تخرجت اليوم بتفوق وحصلت على الوظيفة التي أحلم بها!"
الإجابة: فرح

النص: "لا أصدق حجم الدمار الذي خلفه الإعصار، فقدنا كل شيء."
الإجابة: حزن

النص: "كيف تجرؤ على أخذ أغراضي دون إذن مني؟"
الإجابة: غضب

النص: "سمعت صوتاً غريباً في منتصف الليل والبيت مظلم تماماً."
الإجابة: خوف

5.4 Evaluation Methodology

- **Test set:** 100 sentences evaluated per model per strategy (200 total inferences per model).
- **Temperature:** Set to 0 for all models (deterministic output).

- **Max tokens:** 10 (the model should respond with a single word).
- **Matching:** Model output is compared against both the Arabic and English gold labels after lowercasing and stripping whitespace.
- **Error handling:** API models use exponential backoff for rate limits. Fanar's content filter blocks are detected and marked as "FILTERED" without retrying.

5.4.1 Metrics

We report two standard metrics:

- **Accuracy:** Fraction of correctly classified sentences:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\}$$

- **Macro F1-Score:** Per-class F1 averaged across all 6 emotion classes:

$$F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} \frac{2P_cR_c}{P_c + R_c}$$

  where P_c and R_c are precision and recall for class c, and $C = 6$.

5.5 Results

5.5.1 Overall Performance

Table 7 presents the overall accuracy and macro F1-Score for all five models under both prompting strategies.

Table 7: Overall accuracy and macro F1-Score across all models

Model	ZS Accuracy	FS Accuracy	ZS Macro F1	FS Macro F1
Gemini 2.5 Flash	73.0%	73.0%	58.3%	58.6%
GPT-4o-mini	74.0%	69.0%	46.8%	48.5%
Fanar-C-2-27B	69.0%	62.0%	39.4%	37.8%
Jais-2-8B-Chat	66.0%	66.0%	42.0%	37.9%
ALLaM-7B	61.0%	64.0%	28.3%	30.5%

5.5.2 Comparative Analysis

- **GPT-4o-mini** achieves the highest zero-shot accuracy (74.0%) but its few-shot performance drops to 69.0%, suggesting that in-context examples may introduce a slight bias for this model.
- **Gemini 2.5 Flash** shows the most consistent performance between zero-shot and few-shot (73.0% for both), and achieves the highest macro F1 scores (58.3%/58.6%). This indicates the best balance across all emotion classes.
- **Fanar-C-2-27B** scores lower partly due to its content safety filter, which permanently blocks certain Arabic texts (especially those about violence or strong negative emotions). These blocked responses are marked as "FILTERED" and count as misclassifications.

- **Jais-2-8B-Chat** achieves 66.0% accuracy in both modes, placing it between the API models and ALLaM. Despite having only 8B parameters, it outperforms the 27B Fanar in F1 for zero-shot (42.0% vs 39.4%).
- **ALLaM-7B** has the lowest overall scores (ZS: 61.0%, FS: 64.0%). However, it is the only model that shows a notable *improvement* from zero-shot to few-shot (+3.0% accuracy), suggesting it benefits from in-context examples.

5.5.3 API vs. Open-Weights Models

A clear performance gap exists between API-based models and open-weights models:

- API models (Gemini, GPT, Fanar) average 70.0% zero-shot accuracy.
- Open-weights models (ALLaM, Jais) average 61.5% zero-shot accuracy.

This gap is expected given the significantly larger scale and extensive training of commercial API models. However, the open-weights models offer deployment advantages: full data privacy, no API costs, and the ability to fine-tune for specific Arabic NLP tasks.

5.6 Error Analysis

Common error patterns observed across all five models:

Table 8: Common misclassification patterns

True Label	Predicted	Observation
Surprise (اندهاش)	Fear / Joy	Surprise is the hardest class; models confuse it with fear (negative surprise) or joy (positive surprise)
Anger (غضب)	Sadness / Disgust	Negative emotions are often conflated, especially in dialectal Arabic
Disgust (اشمئزاز)	Anger	Disgust and anger share overlapping sentiment in Arabic expressions

5.6.1 Arabic-Specific Challenges

- **Label mismatch:** The dataset uses اندهاش (astonishment) for Surprise, but models often output مفاجأة (surprise). Both are valid Arabic translations, but only exact matches count as correct.
- **Dialect confusion:** Dialectal Arabic (e.g., Egyptian مش مصدق) is consistently harder for all models compared to MSA or Classical Arabic.
- **Content filtering:** Fanar's safety filter blocks some emotionally charged Arabic texts (particularly those involving violence, fear, or anger), resulting in "FILTERED" outputs that reduce its effective accuracy.
- **Verbose outputs:** Open-weights models (ALLaM, Jais) sometimes generate multi-word answers instead of the requested single-word emotion label, causing match failures despite identifying the correct emotion within the output.

5.7 Manual Adjudication of Predictions

An automated exact-match evaluation can sometimes be overly strict. To ensure evaluation fairness and accuracy, we introduced a manual adjudication step to review the mismatched predictions. This additional review was necessary for three primary reasons:

1. **Label Synonymy:** Models often predict semantically equivalent classifications that differ only in vocabulary. For example, predicting مفاجأة (surprise) instead of the golden label اندهاش (astonishment).
2. **Emotion Overlap:** Many sentences can be validly classified with multiple emotions depending on the interpretation. For instance, sadness and anger may be very similar in certain situations, making the model's alternative prediction perfectly acceptable.
3. **Golden Label Inaccuracies:** Some human errors or subjectivity can occur when originally annotating and collecting the golden labels for the dataset.

To account for these factors, all mismatched predictions across all five models were individually reviewed by a human annotator.

5.7.1 Adjudication Protocol

For each mismatched prediction, the reviewer examined the original Arabic text, the golden label, and the model's output, then classified the mismatch as either:

- **Accepted:** The model's prediction is a valid/acceptable answer based on the reasons outlined above.
- **Rejected:** The model's prediction is genuinely incorrect.

5.7.2 Adjusted Results

After manual review, accepted predictions were reclassified as correct. Table 9 shows the accuracy improvements across all models.

Table 9: Accuracy before and after manual adjudication

Model	ZS Original	ZS Adjusted	ZS Δ	FS Original	FS Adjusted	FS Δ
Gemini 2.5 Flash	73.0%	87.0%	+14.0%	73.0%	86.0%	+13.0%
GPT-4o-mini	74.0%	91.0%	+17.0%	69.0%	84.0%	+15.0%
Fanar-C-2-27B	69.0%	86.0%	+17.0%	62.0%	75.0%	+13.0%
Jais-2-8B-Chat	66.0%	72.0%	+6.0%	66.0%	75.0%	+9.0%
ALLaM-7B	61.0%	69.0%	+8.0%	64.0%	78.0%	+14.0%

INSIGHT

Summary:

✔ Manual adjudication significantly improved the accuracy scores of all models.

✔ Notably, GPT-4o-mini achieved an outstanding 91.0% adjusted accuracy in zero-shot mode.

Key Takeaway: Many apparent "errors" by LLMs in Arabic NLP are actually **acceptable alternative formulations**, not true failures.

5.8 Limitations

While this benchmark provides valuable insights, several limitations must be acknowledged:

- **Dataset Size:** The evaluation was conducted on a relatively small, curated dataset of 100 sentences. While carefully balanced across dialects, a larger dataset would yield more statistically robust findings.
- **Subjectivity in Emotion:** Emotion detection is inherently subjective. Despite manual adjudication, some golden labels might be debated among different human annotators. Future iterations should involve multiple human annotators and calculate inter-annotator agreement (e.g., Cohen's Kappa).
- **Prompt Sensitivity:** LLMs are highly sensitive to prompt phrasing. The simple, standard prompts used (Zero-Shot and 3-Shot) might not elicit the maximum capability of every model, particularly for instruction-tuned models like Fanar that might require specific system prompt formats.

5.9 Final Conclusions and Recommendations

This benchmark evaluated five LLMs on Arabic emotion detection, a task that challenges models with linguistic diversity (MSA, Classical, Dialects) and nuanced emotion boundaries.

Final Conclusions:

1. **Proprietary vs. Open-Weights:** Proprietary models (**GPT-4o-mini** and **Gemini 2.5 Flash**) maintain a noticeable edge in zero-shot accuracy, particularly in handling dialectal nuances and adhering strictly to output formats. However, open-weights models like **ALLaM-7B** and **Jais-2-8B-Chat** demonstrate significant promise and competitive capability, especially when guided by few-shot examples.
2. **Impact of Few-Shot Prompting:** Interestingly, few-shot prompting did not universally improve performance across all models. While it significantly boosted ALLaM-7B (+7.0% raw accuracy), it actually caused a slight regression in GPT-4o-mini. This suggests that highly capable models might overfit to the specific few-shot examples, whereas smaller models use them to understand the task schema.
3. **The Necessity of Adjudication:** The manual adjudication process revealed that a large portion of apparent "misclassifications" by LLMs in Arabic are actually valid synonymous predictions (e.g., مفاجأة vs. اندهاش). Exact-match evaluation severely penalizes models for linguistic richness.

Recommendations:

- **Model Selection:** For zero-shot applications requiring high accuracy and out-of-the-box dialectal understanding, **GPT-4o-mini** (91.0% adjusted accuracy) is highly recommended. For environments where data privacy is paramount requiring an open-weights model, **ALLaM-7B** deployed with few-shot prompting is the most robust choice.
- **Evaluation Methodology:** Standard exact-match evaluation is fundamentally inadequate for generative LLMs. Future Arabic NLP benchmarks **must** incorporate semantic similarity metrics (using embeddings) or an "LLM-as-a-Judge" framework to automate the adjudication process fairly and account for synonymy.
- **Prompt Engineering for Constrained Outputs:** To mitigate the verbosity of open-weights models (which frequently output full sentences instead of the requested single-word label), future workflows should employ strict structural enforcement, such as JSON-schema generation constraints, to ensure parsing reliability.

A Appendix A: Evaluation Queries for al-Ghazali's Philosophy

Listing 1: Evaluation queries for al-Ghazali's philosophy

```
EVALUATION_QUERIES = [
    # Direct / Easy
    {
        "id": 1,
        "query": "              ",
        "type": "direct/easy",
        "description": "Factual question about al-Ghazali's identity and title"
    },
    {
        "id": 2,
        "query": "                ",
        "type": "direct/easy",
        "description": "Direct quote/definition question"
    },
    # Direct / Medium
    {
        "id": 3,
        "query": "                     ",
        "type": "direct/medium",
        "description": "Core philosophical stance"
    },
    {
        "id": 4,
        "query": "                          ",
        "type": "direct/medium",
        "description": "Comparative philosophical question"
    },
    {
        "id": 5,
        "query": "                        ",
        "type": "direct/medium",
        "description": "Causality and science"
    },
    # Indirect / Medium
    {
        "id": 6,
        "query": "                        ",
        "type": "indirect/medium",
        "description": "Reasoning about his superiority"
    },
    {
        "id": 7,
        "query": "                         ",
        "type": "indirect/medium",
        "description": "Relationship between mysticism and philosophy"
    },
    # Direct / Hard
    {
        "id": 8,
        "query": "                        ",
        "type": "direct/hard",
        "description": "Complex theological/philosophical argument"
    },
    # Indirect / Hard
    {
        "id": 9,
```

```
        "query": "                                  ",
        "type": "indirect/hard",
        "description": "Deep interpretive question about asceticism as
            epistemology"
    },
    {
        "id": 10,
        "query": "                                      ",
        "type": "indirect/hard",
        "description": "Abstract distinction between modes of thought"
    },
]
```